

Internally Organized Activity During Offline Brain States

Alice remarked. “I can't remember things before they happen.”

“It's a poor sort of memory that only works backwards,” the Queen remarked.

—LEWIS CARROLL¹

To think is to forget a difference, to generalize, to abstract . . . to sleep is to be abstracted from the world.

—JORGE LUIS BORGES²

When social animals are gathered together in groups, they become qualitatively different creatures from what they were when alone or in pairs. Single locusts are quiet, meditative, sessile things, but when locusts are added to other locusts, they become excited, change color, undergo spectacular endocrine revisions, and intensify their activity until, when there are enough of them packed shoulder to shoulder, they vibrate and hum with the energy of a jet airliner and take off.

—LEWIS THOMAS³

1. *Through the Looking-Glass*, by Lewis Carroll, chapter 5. [NeuroNote: Lewis Carroll had relatively frequent migraines and at least two documented epileptic seizures. His own experience with seizures may have informed Alice's experiences in *Alice in Wonderland* (Woolf, 2010)].

2. The quote is from Borges' “Funes, the Memorious” (1994). [NeuroNote: The piece was written, according to Borges, as a means of staving off his insomnia (Borges and Dembo, 1970).]

3. Thomas (1972).

Lewis Thomas's beautiful metaphor of population cooperativity could equally well describe a peculiar hippocampal population pattern, called sharp wave ripples—just substitute neurons for locusts. The sharp wave ripple is a randomly emerging local field potential event occurring when many neurons emerge from their sessile state and fire together shoulder to shoulder. I have been enthralled by their beauty and power from the first moment I heard their buzzing sound in my postdoctoral years. I felt as if I had been listening to a group of orchestra musicians idly tuning their instruments, and the next moment they united in the thrilling harmonies of Beethoven's *Fifth Symphony*. I still think it is the most beautiful pattern the brain produces. Sharp wave ripples represent among the most synchronous population patterns in the mammalian brain, more synchronous than the responses evoked by sensory stimulation of any strength. Yet they are self-organized and spontaneously emitted by hippocampal circuits. The brain must have a good reason to go to the trouble of producing such unusual population activity, which is totally unexpected from the outside-in view of the brain. My first thought was that the sharp wave is an abnormal epileptic activity, perhaps triggered by mechanical trauma inflicted by the recording electrodes I placed into the brain. After numerous control experiments, it became clear that, instead, they represented a potentially important physiological pattern, one whose mysteries waited to be uncovered.

There is no trigger for the occurrence of sharp wave ripples. They are not caused by anything. Instead, they are released, so to speak, when subcortical neurotransmitters reduce their grip on hippocampal networks, as routinely happens during nonaroused or idle waking states, such as sitting still, drinking, eating, grooming, and non-rapid eye movement (REM) sleep.⁴ Sharp wave ripples are produced by tens of thousands of neurons in the hippocampus, subicular complex, and entorhinal cortex, firing over just 30–100 ms at times when the brain is disengaged from the environment.⁵ Any more synchrony than that could trigger an epileptic seizure. This neuronal cooperation is what fascinated me most. Synchrony can bring about enormous power surges at no cost. If a symphony is played one instrument at a time, the cooperative action of many instruments can never be identified. Similarly, when we listen to the spike

4. Sharp wave ripples are the main population pattern seen in the isolated and transplanted hippocampus (Buzsáki et al., 1987). Conversely, when septo-hippocampal cholinergic neurons are activated by optogenetic methods, the synaptically released acetylcholine prevents the occurrence of sharp wave ripples (Vandecasteele et al., 2014).

5. Occasionally, multiple ripples occur in a sequence or fuse together and reflect exceptionally long neuronal assembly sequences (up to several hundred milliseconds). These long sequences most often occur in novel environments and memory-demanding conditions.

sounds of individual neurons (over a speaker that allows researchers to monitor the signals coming from their electrodes), we cannot detect a sharp wave ripple, no matter how long we listen to neurons one by one. But when many neurons emit action potentials together, their collective action sounds like a jet airliner. I wondered what the function of this collective action might be.

In the 1980s, many laboratories were searching for natural conditions that could mimic artificially induced synaptic changes in neurons in a dish, used as a model for memory.⁶ And there it was. The sharp wave ripple shared numerous characteristics of the electric pulse patterns that brought about long-term changes in synaptic plasticity, such as duration, high frequency, and the super powerful synchrony required for a teaching signal.⁷ Sharp wave ripples are found in every mammalian brain investigated to date in the same form and shape—and may serve identical functions (Figure 8.1). They form discrete quanta of spike sequence packages. As early as my doctoral dissertation, I made the bold claim that hippocampal sharp wave patterns represent the sought-after biological marker for information compression and plasticity. Then I spent the next three decades trying to understand their secrets. During these years, the views about the sharp wave ripple have enriched considerably. This peculiar and unique brain pattern is viewed today as a subconscious mechanism to explore the organism's options, searching for stored items of the past in the disengaged brain in order to extrapolate and predict possible future outcomes. It embodies a brain mechanism that compresses the discrete concepts of past and future into a continuous stream.

EXCITATORY GAIN DURING SHARP WAVE RIPPLES FACILITATES SPIKE TRANSMISSION

The sharp wave ripple is a cascade of two events, the sharp wave and the ripple, which occurs relatively simultaneously but at different locations (Figure 8.2). Sharp waves are large-amplitude, negative-polarity deflections in the apical

6. *Synaptic plasticity* is often used synonymously with *memory*. However, alternation of circuits could be also brought about by changing the intrinsic properties of neurons and axon calibers without affecting synaptic number or strength.

7. At an early meeting on the physiology of the hippocampus, Tim Bliss, the discover of long-term potentiation, encouraged me after my talk in front of a skeptical audience: "My point about sharp wave was that it is a naturally occurring phenomenon which satisfies the two conditions, high frequency and high strength" (p. 395 in Buzsáki and Vanderwolf, 1985). I have recently written a monograph on sharp wave ripples for my specialist neuroscientist colleagues. Mechanisms, implications for memory, planning, disease, historical details, and the pertinent references can be found there (Buzsáki, 2015).

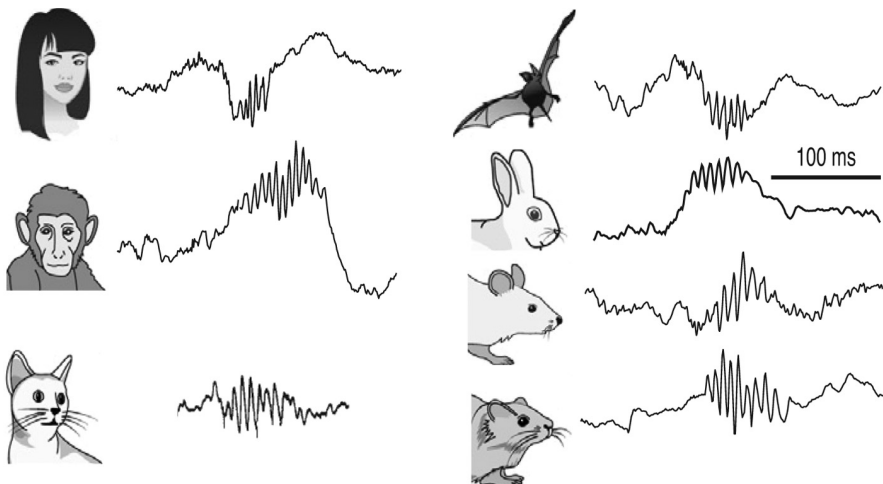


Figure 8.1. Preservation of sharp wave ripples in mammals. Illustrative traces recorded from various species.

Reproduced from Buzsaki et al. (2013).

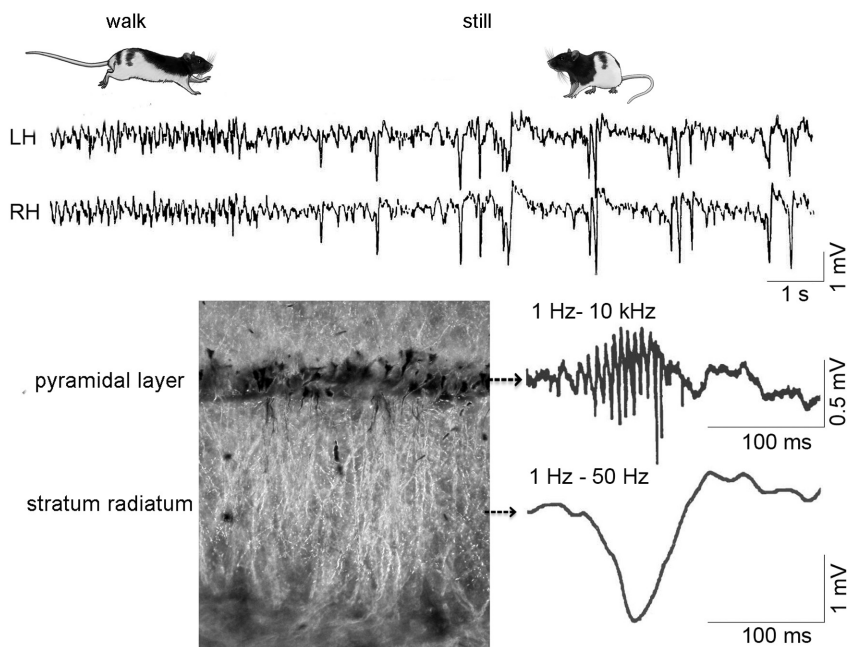


Figure 8.2. Behavior-dependence of hippocampal local field activity. *Top:* Extracellularly recorded events from symmetric locations in the dendritic layers of the left (LH) and right (RH) dorsal hippocampus during locomotion–immobility transition. Note regular theta waves during locomotion and large amplitude, bilaterally synchronous negative waves (sharp waves, SPW) during immobility. *Bottom:* A single sharp wave recorded from the dendritic layer (red) and simultaneously occurring ripple event recorded from the pyramidal layer.

dendritic layer of CA1 pyramidal neurons, where most of the incoming synaptic contacts are made by the axons of their CA3 partners. The sharp wave is a telltale of strong firing coordination among many neurons in the upstream CA2 and CA3 regions. Measuring the sharp wave amplitude is, therefore, a convenient way to quantify the magnitude of this synchrony. The self-organized population activity underlying sharp waves emerges from the strong excitatory influence of the extensive recurrent collaterals of the CA3 pyramidal neurons.⁸ In turn, the strong depolarization of CA1 pyramidal neurons and interneurons brought about by the synchronous activity of the CA3 neurons produces a tug of war between excitatory and inhibitory neurons, resulting in a short-lived “ripple,” a fast oscillatory synchronization (140–200/s) of these two competing populations (Figure 8.2).⁹ The pyramidal neurons of the CA1 region provide the only hippocampal output to cortical targets, so this activity must have functional significance.¹⁰ We just have to figure out what it is.

The most interesting part is the neuronal dynamic that evolves around these events. Nearly all neuron types participate in the ripple events. Some inhibitory interneuron types remain neutral or decrease their activity after a transient increase, but the majority robustly increase their firing. The overall result is that while both inhibition and excitation increase during sharp wave ripples, excitation makes a two- to threefold gain over inhibition.¹¹ This is the highest gain

8. Prior to the full organization of the large recurrent excitatory CA3 system, sharp wave bursts are already present in the postnatal brain. However, nearly every event is driven by some kind of movement from the body (Leinekugel et al., 2002). The proprioceptive reafferentation drives neocortical–entorhinal circuits which, in turn, trigger hippocampal sharp wave bursts. By the second week of life, the hippocampus disengages from its dependence on body-derived signals and generates self-organized sharp waves (Valeeva et al., 2018). Ripples in CA1 emerge by the third week of postnatal life, coinciding with the sequential firing of hippocampal neurons (Buhl and Buzsáki 2005).

9. Buzsáki et al. (1992). Ripple patterns occur in many physical systems when a large number of random, independent events are combined. The most famous such event is the Big Bang, where many random events are assumed to have generated stochastic gravitational waves. These gravitational waves, also called ripples, are approximately 250 Hz and last for 100 ms, reflecting two massive bodies spiraling into each other (Cho, 2016).

10. This is not quite correct, though, as a small fraction of hippocampal inhibitory interneurons also send their axons to several other extrahippocampal regions (Jinno et al., 2007). We called this special type of cell a “long-range” interneuron (Buzsáki et al., 2004) in reference to the “short-cut” connections of small-world network architecture (Watts and Strogatz, 1998).

11. Read more about gain in Chapter 11. The large gain may derive, paradoxically, from the highly reliable spike transfer between pyramidal cells and interneurons (Csicsvari et al., 1998; English et al., 2017). Because pyramidal neurons fire highly synchronously in a given ripple wave and because even a single pyramidal neuron spike can discharge fast spiking interneurons, spiking of many additional pyramidal cells within the spike refractory period of the interneuron

of excitation in the hippocampal system, making the sharp wave event especially suitable to transmit information from the hippocampus to the neocortex (see Chapter 11). But what kind of neuronal information is transmitted? After all, sharp wave ripples are self-organized events that emerge in the “offline” or idling states of the brain. It seems odd for the brain to be so active at times when no perceptual inputs bombard its sensors.

ORGANIZED SPIKE SEQUENCES OF SHARP WAVE RIPPLES

What makes hippocampal sharp wave ripples so special, in addition to their slender oscillatory patterns and strong excitatory gain, is the way that neurons emit their spikes relative to each other. They can play palindrome, a sequence of letters that spell the same words whether read forward or backward. “Was it a car or a cat I saw?” This palindromic role makes sharp wave ripples suspicious of some coding mechanism because similar tricks have been exploited in molecular evolution by retroviruses. A retrovirus can enter the cell nucleus and patch its own DNA into the cell’s DNA, adding new information to an existing knowledge base, which then becomes permanent. A palindromic repeat of DNA coding would look like GTTCCTAATGTA–ATGTAATCCTTG.¹² Sharp wave ripples can also do such things, and the ability to switch the direction of the trajectory must also have special importance for neuronal computation.

Recall from Chapter 7 that, during exploration, neuronal sequences occur at both the theta time scale and the time scale of behavior, and that these sequences are specific to the animal’s path. For example, in Figure 8.3 place cells 1–13 are active in different segments of the maze track. At any given time on the track, only a small fraction of the neurons fire together within the theta waves. However, at the beginning and the end of the track, where the rat is sitting immobile, virtually all neurons fire together. When we zoom in and examine the firing patterns of these same neurons at an expanded time scale, we find that they also show a sequence. The sequences occur either in the same order as the sequential firing on the track or in the reversed order, but at an accelerated pace in both cases, and they coincide with fast ripple oscillations in

(1–2 ms) cannot recruit more spikes and more inhibition. This refractoriness may be the key mechanisms for the large excitatory gain during sharp wave ripples (Csicsvari et al., 1999).

12. Work on the special coding abilities of retroviruses paved the way to today’s CRISPR magic of molecular biology (Quammen, 2108). CRISPR, which stands for *clustered regularly interspaced short palindromic repeats*, sounds like spike patterns of sharp wave ripples.

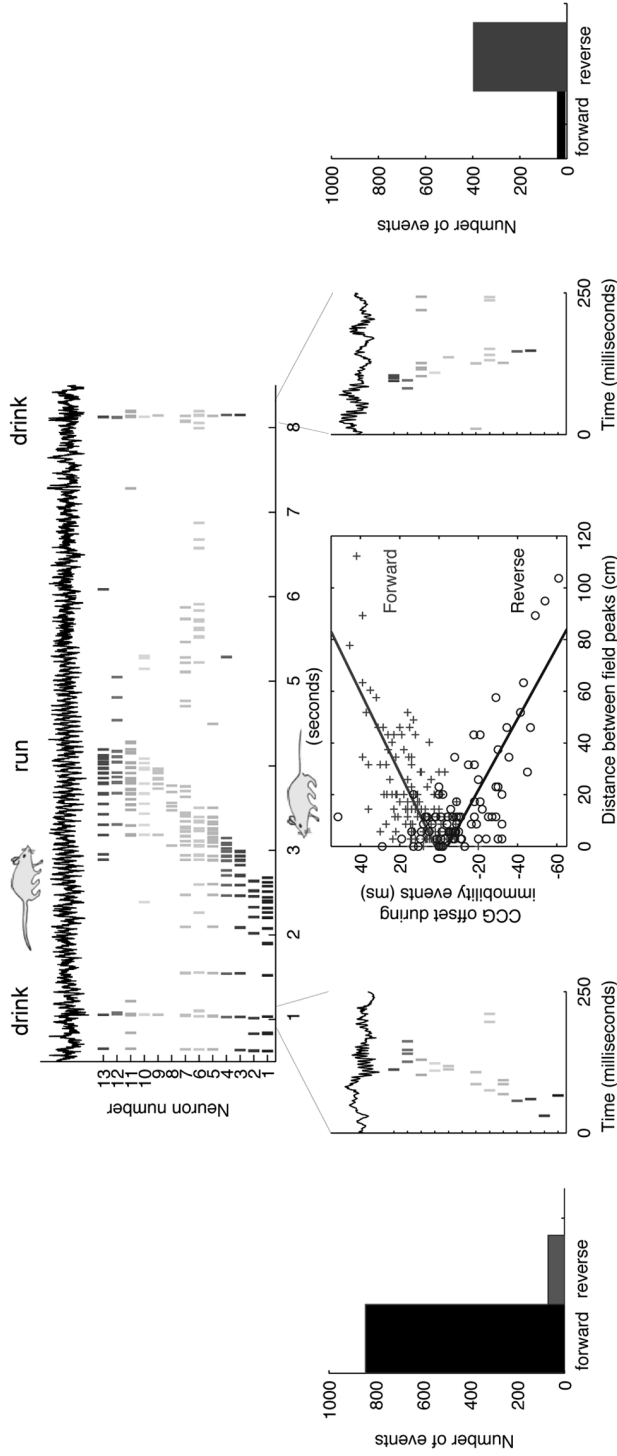


Figure 8.3. Forward and reverse replay of place cell sequences. *Top:* Spike trains of thirteen neurons before, during, and after a single lap (*top continuous trace*: CA1 local field potential). The animal runs from the left to the right ends of the track and collects water reward at both ends. The insets below magnify 250-msec sections of the spike train, depicting synchronous forward and reverse replay, respectively. Bar graphs (bottom left and right) show that 95% of forward events took place before the run, whereas 85% of reverse replay events occurred after the run. *Middle:* Temporal offsets (CCG) between spikes of neuron pairs during the synchronous forward (+) and reverse (o) replay events are correlated with distance representations between place field peaks on the track. From Diba and Buzsaki (2007).

the field recordings. It is as if the brain played a movie back for us on fast forward or reverse that recapitulated the motion of the rat in the maze. Without further analyses, we can only wonder what functions these accelerated replay events may serve.

The majority of fast-forward events occur at the start of the track, and their incidence steadily rises until the run begins as if the hippocampus rehearses or plans the future travel path. In contrast, reversed replay events dominate at the other end of the track, after the run as if the brain recapitulated the path in a reversed order. The preponderance of reverse trajectories can be manipulated by the magnitude of the reward the animal obtains at the end of the journey.¹³ In short, the sharp wave ripple is a compression of event sequences that have either already occurred or will evolve in the future. Beyond faithfully preserving sequential order, both forward and reversed events during sharp wave ripples also contain information about the distances between place fields on the track (Figure 8.3). Recall that this distance-to-time conversion is reminiscent of the theta time scale compression of place cell distances discussed in Chapter 7, with two notable differences. First, the time compression factor is typically larger during sharp wave ripples than during theta oscillations.¹⁴ Second, during theta oscillations, only forward sequences are observed, whereas the neuronal movie of the animal's journey in the maze can be seen in both forward and reverse order during sharp wave ripples. Importantly, the fast replay events during ripples are not assisted by external signals or body-derived cues. Compressed

13. Typically, many more neurons participate in sharp wave ripples than the number of place cells in a maze. This makes it complicated to compare real-time sequences with replay sequences. Foster and Wilson (2006) introduced a novel and effective method by regarding place cell sequences in the maze as a template and searching for a similar rank order of spikes during sharp wave ripples. They observed only reversed replay events on the track, in support of a previous prediction (Buzsáki, 1989), and contrasted them with the assumption that only forward sequences occur during sleep. According to Foster and Wilson, reversed "replay is suggestive of a role in the evaluation of event sequences in the manner of reinforcement learning models." I expressed a similar view: during sharp wave ripples, "recurrent collateral excitation in the CA3 region spreads hierarchically: the most excitable cells fire first followed by less excitable ones, that is in the reversed order to that in which they were potentiated during exploration" (Buzsáki, 1989). Reverse replay can be enhanced by the magnitude of the reward (Ambrose et al., 2016). Compressed replay of learned information has also been observed in humans (Kurth-Nelson et al., 2016).

14. Distance compression is approximately 30% stronger during sharp wave ripples compared to theta waves, commensurate with the stronger network drive during ripples (Diba and Buzsáki, 2007; Drieu et al., 2018). The speed of replay during sharp wave ripples corresponds to 8 m/s, which is 10–20 times faster than the average speed at which the rat runs or walks through those place fields (Davidson et al., 2009). These time scales are determined by internal brain states rather than by sensory inputs.

replay is like the scaled version of a song or speech; it is identified as the same sequential pattern, just played faster.

Although sharp wave ripples are self-organized events, their spike content is related to the firing patterns evolving in real time in the behaving animal. We can conclude that forward events before the run may serve to plan the upcoming locomotion trajectories. Conversely, reversed replay events at the end of the run may recapitulate the trajectory in the maze, so the animal remembers that choice. Thus, the forward and reversed sequences of spontaneously induced sharp wave ripples may serve both prospective and retrospective roles (i.e., prediction and postdiction), just as the Queen noted in Carroll's masterpiece. Based on the discussion in Chapter 5, we can speculate that internalization of travel trajectories can be used both to repeat snippets of past experience and to imagine or plan actions. After all, the only biological utility of memory is to help predict the future.

CONSOLIDATION OF LEARNED EXPERIENCE DURING NON-REM SLEEP

Manipulations that affect synaptic strength distributions in hippocampal networks can have a profound impact on the neuronal membership of spontaneous network events such as sharp wave ripples. Artificially altering synaptic strengths between CA3–CA3 or CA3–CA1 neurons can alter the waveform of sharp wave ripples, make new place fields, or make place fields disappear. These findings suggest that the neuronal composition of sharp wave ripples can be biased by the recent past of the hippocampal network. Once a new sharp wave–associated neuronal sequence emerges, the sequence repeats spontaneously many times after the initial condition that induced it. In this way, firing patterns of the entorhinal inputs can modify intrahippocampal synaptic connections, and these modifications are reflected in spontaneous, self-organized population events.¹⁵ Let's pause here briefly to appreciate what good service the sequence of neuronal events can do for the hippocampus and its targets: the synaptic-cellular modification of neurons brought about during learning can be read out later from the spontaneously emerging neuronal activity. This sequential

15. Manipulations that affect synaptic strength distributions in hippocampal networks have a profound impact on the neuronal membership of spontaneous network events. Electric stimulation of different sets of afferent entorhinal axons can induce a unique spatial-temporal constellation of evoked responses in the hippocampus. After tetanic stimulation (trains that induce long-term potentiation of synapses), the modified connections are reflected in the spontaneous, self-organized population events, so that the spontaneously emerging events resemble the evoked responses (Buzsáki, 1989).

mechanism may be regarded as the physiological foundation for a two-stage model of memory formation.

The hypothesized consolidation model works in two steps. During learning, associated with the theta rhythm, afferent activity from the neocortex–entorhinal cortex pathway brings about a transient change of synaptic strengths in the CA3 hippocampal regions and within CA3–CA1 connections, where the learned information is temporarily held. This first stage is followed by a consolidation phase during which the same neurons and synapses that were active in the learning process are repeatedly reactivated during sharp wave ripples. Because hundreds to thousands of sharp wave ripples occur during non-REM sleep, I hypothesized that they are critical for consolidating labile memories into a more permanent form. Sharp wave ripples repeat snippets of the previously learned information over and over during sleep, after the experience. This protracted process fits nicely with research on episodic memories. They are not instantaneous veridical pictures of a situation, like a picture taken by a digital camera. Instead, episodic memories resemble Polaroid pictures, in which an initially faint scene gradually becomes crispier. During the process, the picture is vulnerable to modification, as are our memories.

The first convincing evidence to show replay of learned experience came from a groundbreaking study by Matt Wilson and Bruce McNaughton at the University of Arizona at Tucson.¹⁶ They examined the co-occurrence of spikes of hippocampal CA1 pyramidal neuron pairs in 100-ms windows, which they called “coactivations,” during sleep both before and after behavioral experience on a familiar open-field foraging task. As expected, the activity of neurons with overlapping place fields showed a highly positive correlation during exploration, whereas pairs with nonoverlapping fields showed no reliable temporal correlations. Critically, neuron pairs with overlapping place fields and strong correlations during foraging continued to fire together more strongly during sleep afterward, although they had been less correlated during sleep before foraging. Numerous experiments in many laboratories have supported and

16. Wilson and McNaughton (1994); Kudrimoti et al. (1999); Skaggs and McNaughton (1996); Hirase et al. (2001). By and large, these physiological findings support previous ideas, based on experiments with hippocampus-damaged patients, that the initially labile memory trace becomes consolidated over time. However, while the psychological experiments promoted the view that the initially hippocampus-dependent memories are transferred to the neocortex and become independent of the hippocampus (Scoville and Milner, 1957; Squire, 1992a, 1992b; Squire et al. (1975); Ferbinteanu et al., 2006; Squire and Alvarez, 1995), an alternative view is that consolidation corresponds to preservation of a crude “index” of neocortical memories in the hippocampus (Teyler and DiScenna, 1986; Nadel and Moscovich, 1997). In this way, hippocampal sequences can sweep through representations of detailed neocortical semantic information and concatenate them into a virtual offline episode (Buzsáki, 2015).

expanded these findings. One important extension showed that those neuronal trajectories (similar to those seen in Figure 8.3) reactivated many times during the sharp wave ripples of non-REM sleep are an accelerated version of place cell sequences during exploration.¹⁷ Thus, replay of learned information has been demonstrated.¹⁸

A Revised Version of the Two-Stage Model

These early speculations and experiments supported the idea that, during learning, large-scale synaptic modifications occur in the hippocampus, and the transiently modified synapses get stronger or at least persist (i.e., consolidate) during offline states of the brain. In turn, learning-induced synaptic changes are believed to entirely determine how neurons organize themselves into sequences during sharp wave ripples. In a sense, this description of events can be interpreted to mean that novel experience builds up a new neuronal trajectory from scratch, in line with the outside-in view. Yet the same findings and the experiments we are going to discuss herein are also compatible with the alternative, inside-out view that we discussed in Chapter 7: namely, that

17. Nádasdy et al. (1999); Lee and Wilson (2002) introduced a behavioral event-based template-matching method where the sequences of smoothed place fields of hippocampal pyramidal cells during exploration defined the template (a “neuronal word”). Higher order relationships are important for making “associative inference.” For example, if *A* is related to *B*, and *B* is related to *C*, then *A* is indirectly related to *C* (in a higher ordered manner, mediated by the hidden variable *B*). Such inference depends on the integrity of the hippocampus (Dusek and Eichenbaum, 1997; Schacter and Addis, 2007; Schacter et al., 2007). In all these early experiments, reactivation of waking sequences was in the forward direction, largely because these were the only templates examined. When researchers looked for reverse sequences, they were also found during sleep (Wikenheiser and Redish, 2013), thus demonstrating that reverse replay is not specific to sharp wave ripples during waking states (Foster and Wilson, 2006).

18. These ideas did not remain unchallenged. Lubenov and Siapas (2008) suggested that because replaying waking experience involves a small fraction of neurons, neuronal firing during sharp wave ripples reflects a transition state between randomness and synchrony. They suggest that the synchronous bursts can lead to long-term depression of synapses and the selective erasure of information from hippocampal circuits as memories are transferred to neocortical areas. However, memories do not cease to exist in the hippocampus (Nadel and Moscovitch, 1997). The potential role of sharp wave ripples in sleep homeostasis is discussed in Grosmark et al. (2012). Foster (2017) questions the role of waking sharp wave ripples in memory and, instead, emphasizes their prospective role. However, this view leaves the coexistence of both forward and reverse replays during both sleep and waking unexplained. Norimoto et al. (2018) suggested that sharp wave ripples protect experience-active neurons while downscale memory-irrelevant neuronal activity.

the hippocampus can generate myriads of sequences internally even without experience and that various aspects of learning can be linked to some of these preexisting sequence patterns.

According to this revised view, the first stage of the two-stage model is a selection process in which one or a few of the existing sequences or their concatenated variants are matched to the experience. Because every novel situation has elements of familiarity (Chapters 12 and 13), the brain can select a neuronal trajectory that best describes the current situation. The brain always guesses; that is, it attempts to deal with any event even in the most unexpected situation by matching it with a particular existing neuronal pattern. If needed, new neurons can be added (or deleted) to this selected backbone to refine the sequence adapted to the novel aspects of the situation. In turn, these new neurons are incorporated (or eliminated) in the selected sequence during sharp wave ripples (stage 2). While the selection-incorporation version of the two-stage model appears to differ from the previous consolidation version only in small details, it is more economical. Only small changes are expected to occur in the hippocampal and neocortical networks because a large part of learning is a selection process—matching of the experience to preexisting patterns (Chapter 7) rather than a new construction.¹⁹ To develop this argument further, we need to learn important details about the statistics of neuronal dynamics, which we discuss in Chapter 12, and we revisit the selection hypothesis in Chapter 13.

Furthermore, the experiments on sharp wave ripples discussed thus far only demonstrate that reactivated neuronal patterns of waking experience during sharp wave ripples may do something useful in the hippocampus, but not beyond. If the occurrence of sharp wave ripple-associated replay is irregular, rather than predictable, then how can the neocortex read the messages reliably without being on the alert all the time? As discussed earlier, the neocortex reboots itself from complete silence of neuronal activity (“down” states) hundreds to thousands of times during non-REM sleep. If replay from the hippocampus occurs during neocortical silence, it might fall on deaf ears unless hippocampal and cortical activities are somehow coordinated. Fortunately, they are.

19. Admittedly, these dense paragraphs are based largely on speculation, derived mainly from theoretical considerations and, to a smaller extent, on recent experiments (Luczak et al., 2009; Dragoi and Tonegawa, 2011, 2013a, 2013b; Mizuseki and Buzsáki, 2013; Buzsáki and Mizuseki, 2014; Grosmark and Buzsáki, 2016; Liu et al., 2018). I expand on these ideas in Chapter 13.

Coupling Sharp Wave Ripples with Neocortical Events

The two major neocortical events during non-REM sleep, slow oscillations and sleep spindles, have a nuanced relationship with hippocampal sharp wave ripples. The direction of communication may depend on sleep state, target structure, and previous experience. The strongly synchronous hippocampal output during sharp wave ripples can latch prefrontal circuits to the down state.²⁰ In the return direction, when neocortical spindles successfully invade entorhinal–hippocampal networks, they can time sharp wave ripples. These triggered events are likely important because the punctate hippocampal output can further enhance the activity of the same still-active neurons that initiated the neocortical–hippocampal dialogue.²¹ Although the hippocampus receives disparate inputs during successive cortical events, this process allows it to find or index the active cortical patches. Thus, the interplay of hippocampal sharp wave ripples and neocortical ripples creates temporal windows that facilitate the exchange of information.

These correlational experiments have been supplemented by experimental manipulation of sharp wave ripple content. In one clever experiment, the spiking activity of place cells during sleep was coupled with rewarding brain stimulation. Stimulation of many sites in the brain feels good and could be used experimentally as a convenient substitute for reward. Stimulation of the chosen rewarding brain site was triggered by the spontaneous emission of each spike of a place cell during sleep ripples. When the mouse awoke, it made a beeline for the location represented by that place cell as if searching for the reward. The stimulation formed a new relationship between reward

20. Peyrache et al., 2011. Interictal spikes in the epileptic hippocampus may represent super-strong, pathological versions of sharp wave ripples and consistently induce down–up transitions and spindles in the prefrontal cortex even in the waking brain. The abnormal epileptic spikes broadcast “nonsense” messages to the neocortex and induce spindles at times when the prefrontal areas are not in the “ready” state. Thus, all three suspected patterns of memory consolidation—hippocampal ripples, neocortical slow oscillations, and spindles (Diekelmann and Born, 2010)—are aberrant in the epileptic brain and can be responsible for memory/cognitive impairment in chronic epileptic patients (Gelinas et al., 2016).

21. Siapas and Wilson (1998); Sirota et al. (2003); Isomura et al. (2006); Mölle et al. (2009); Johnson et al. (2010); Sullivan et al. (2014). A few experiments examined joint replay of waking experience in the hippocampus and neocortical structures (Ji and Wilson, 2007; Peyrache et al., 2011; O’Neill et al., 2017; Khodagholy et al., 2017). Replay in the neocortex also occurs in brief bursts, lasting 100 ms or so, typically coincident with the up-state of slow oscillation (Takehara-Nishiuchi and McNaughton, 2008; Johnson et al., 2010; Rothschild et al., 2017). Most cortical assembly reactivation events occur in concert with hippocampal sharp wave ripples.

and spatial location.²² To complement the role of replay in memory consolidation, another experiment demonstrated that selectively abolishing sharp wave ripples during sleep following learning in a maze impairs spatial memory performance.²³

The two-stage model, using sharp wave ripples as a key mechanism for replay, explains why episodic information can be remembered even after a single experience. Instead of repeating the learning process behaviorally, the endogenously organized sharp wave ripples, perhaps in partnership with thalamocortical spindles and slow oscillations, do the heavy lifting. They repeat various aspects of recent experience many times at a compressed time scale and in a strongly synchronized manner so that the neocortical targets take notice of the powerful hippocampal outputs.²⁴ In short, sharp wave ripples may allow the brain to solidify recently learned material. But they do more than that.

CONSTRUCTIVE FUNCTIONS OF SHARP WAVE RIPPLES

The conclusion that sharp wave ripples strengthen neuronal patterns that occurred before “now” is based on the conceptual distinction of the past and present. However, recent thinking questions the validity of this traditional distinction (see Chapter 10). In line with the idea about the contiguity of postdiction and prediction, experiments paint a richer picture for sharp wave ripples, especially the waking variant, beyond consolidating memories. One interesting

22. de Lavilléon et al. (2015).

23. Girardeau et al. (2009). The results of Ego-Stengel and Wilson’s (2010) studies in a “wagon wheel” maze further support the contribution of the sharp wave ripples of post-learning sleep to spatial memory. Jadhav et al. (2012) interfered with sharp wave ripples in the waking, rather than sleeping, rat. Truncation of waking sharp wave ripples during task performance increased choice errors in spatial working memory but not in reference components. In addition, selective truncation of sharp waves that occurred in the vicinity of reward destabilized the spatial map (Roux et al., 2017). Finally, increasing the coupling between sharp wave ripples and slow oscillations by electrically inducing down states of cortical activity or spindles by optogenetic means increased memory performance (Maingret et al., 2016; Latchoumane et al., 2017).

24. Phrased differently, a single experience (incidental learning) can be remembered because “the SPW event would in essence compress time and allow temporally distinct neuronal representations to be combined into a coherent whole” and sharp wave ripples repeatedly replay or “reactivate the same subset of neurons in CA3 and CA1 precisely determined by the recent past of the neural network” (Buzsáki et al., 1994; p. 168). Buzsáki (1989, 1996, 1998, 2015); Lee and Wilson (2002); O’Neill et al. (2006, 2008). Dupret et al. (2010); van de Ven et al. (2016).

study used a long maze that was twisted around to fit the recording room. The duration of replay sequences appeared to depict behavioral trajectories at a constant speed; that is, with duration proportional to distance depicted. This explains why the researchers found up to 700-ms sequences, concatenated from multiple ripple events, that wended their way around the bendy track without jumping between physically close (but uncrossable) corridors. For comparison, smaller mazes typically evoke 50–100 ms events. However, most ripple events were shorter and jumped between discontinuous corridors along the length of the maze track. In other words, replay events do not necessarily represent temporally compressed veridical episodes of travel. Some sequences instead represent snippets and fragments of traversable distances, as if evaluating alternative paths while others play out sequences corresponding to trajectories the animal has never traversed, starting and ending at positions different from that of the animal.²⁵ Thus, similar to the egocentric-allocentric divisions of navigation, simulated navigation of fictive paths during sharp wave ripples can be either egocentric (moving to or away from self) or allocentric (ordering events at other places). These experiments, together with the forward replay of upcoming place cell sequences, suggest a prospective or constructive role for sharp wave ripples.

In most early experiments, replay sequences were compared to a template based on the ordered sequence of place cells on the maze tracks. In essence, it is a targeted search for a needle in a haystack. However, this method comes with a cost: it throws out the many other neuronal sequences that occur during sharp wave ripples and so cannot examine their potential role. In a T maze (which has a central arm and two choice arms), only two templates are examined during sharp wave ripples: left and right turns and their corresponding place cell sequences. In such a simple memory task, neuronal sequences related to sharp wave ripples may correspond to either left or right turns, as if the hippocampus contemplates the proper choice.²⁶ However, during natural foraging, many choices are available at each position of travel. Forward replay may serve to evaluate alternative scenarios and calculate optimal choices.

25. Davidson et al. (2009); Wu and Foster (2014); Foster (2017); Karlsson and Frank (2009); Gupta et al. (2010); Liu et al. (2018). Theta sequences can also serve such a “contemplating” function, especially at maze choice points, where sequentially active neuronal assemblies can “look ahead” toward either the right or left arms of a T maze (Johnson and Redish, 2007; Redish, 2016).

26. Singer et al. (2013). In this task, the neuronal trajectories reactivated during sharp wave ripples preceding correct trials were biased toward representing sequences that proceeded away from the animal’s current location.

A laboratory equivalent of foraging is a search for reward in a large open arena with numerous wells. In this task, the rat is trained to find food hidden in a random well and then return to home base for another reward. If the home base is varied from one day to the next, the rat learns to locate the new home base in just a few trials, after which it returns to it from any food well in a few seconds. The main finding of such an experiment was that before the animal returned to the home base from any location in the arena, its sharp wave ripple-associated sequences often closely matched the place cell sequence between the current location and the home base. The forward neuronal sequences in this two-dimensional exploration task better matched the rat's future path than its past path.

These findings, therefore, demonstrate that previously acquired memories can be flexibly manipulated internally to evaluate alternative future paths and plan optimal behavioral trajectories to a remembered goal, even when those trajectories have never been taken before.²⁷ Other experiments suggest that neuronal sequences are often “stitched together” from fragmented representations involving multiple concatenated sharp wave ripples containing both forward and reversed events.

Before closing the discussion on the constructive aspects of sharp wave ripples, it is important to emphasize that replay does faithfully mimic the true sequential order of the experienced past. Instead, recent experience is embedded into an existing network dynamic. A large part of the dynamic is preformed and guided by the brain's genetically determined construction and existing knowledge base (Chapter 12). Thus, replay sequences reflect more the brain's “beliefs” than mirroring the order of world events. We can speculate that this constructive role of sharp wave ripples can assist in weaving together paths and events that relate to each other even if they have not been experienced physically.

27. Pfeiffer and Foster (2013); Papale et al. (2016); Pfeiffer (2017); Pezullo et al. (2017); Liu et al. (2018); Xu et al. (2018). These experiments confirmed previous theoretical predictions (Schmajuk and Thieme, 1992). The hippocampus graph model of Muller et al. (1996) suggested that distances in a two-dimensional map are reflected by the reciprocal value of synaptic strengths between place cells in the CA3 recurrent system. Once the main landmarks of an environment are mapped onto hippocampal assembly representations, all possible combinations (e.g., the route from any location to the home base) can be computed even if those routes were never traversed by the animal. It has been postulated that sharp wave ripples are exploited for this purpose (Muller et al., 1996; Samsonovich and Ascoli, 2005), and possibly the same mechanisms can be used to generate novel solutions to non-navigational problems (Buzsáki and Moser, 2013).

Are Animals Aware of the Content of Sharp Wave Ripples?

Single sharp wave ripples may be too short to induce a conscious experience, which requires as long as half a second.²⁸ Yet stitching together neuronal trajectories from sharp wave ripples can be conceived of as a mechanism for preconscious contemplation. When you let your mind wander off the leash for a while, that may be a better way to reach the correct solution than trying to concentrate. We all have such experiences: a while after we give up on an effortful search for a name, it may just pop up. Sequences predicting future actions or solutions can be constructed by chaining together clusters of sharp wave ripples so that spiking activity in each sharp wave ripple in the cluster corresponds to different potential segments of a potential route.

Overall, the selection of possible forward paths during sharp wave ripples can be conceived as an internalized vicarious trial-and-error process (Chapter 5) that flexibly “imagines” real or fictive alternatives to select an optimal path or construct novel inferences without the need for movement-based exploration.²⁹ Because neuronal assembly sequences in the hippocampus serve not only spatial but numerous nonspatial functions, the compressing-mixing functions of sharp wave ripples can also support covert simulation of many real-world scenarios). Recently acquired and preexisting knowledge can be combined during sharp wave ripple replay to influence choices, to plan actions, and, potentially, to facilitate generalization, abstraction, and creative thought. In support of this outlined scenario, failure to replay the appropriate trajectory sequence before a choice can predict behavioral errors.³⁰

28. Libet (2005) called it “mind time.” This time may be needed to mobilize large resources in multiple brain structures and allow them to communicate effectively.

29. Schmajuk and Thieme (1992), following Tolman (1932), discussed the idea that head-scanning movements at maze choice points may correspond to “vicarious trial-and-error,” contemplative behavior to select optimal travel paths (Tolman, 1932). Artificial intelligence researchers have also been inspired by the two-state model of memory. The celebrated deep Q-network (DQN) can outperform humans in numerous video games by learning to transform a vector of image pixels into a policy for selecting actions (e.g., joystick movements). The network stores a subset of the training data in an instance-based way and then “replays” it offline over and over, learning anew from successes and failures that occurred in the past. This replay is critical to maximizing data efficiency (Hassabis et al., 2017).

30. Ólafsdóttir et al. (2017). These authors also demonstrated that forward and reverse replays of maze trajectories in sharp wave ripples occur within 5 s before the animal starts the journey or arrives at the goal location. If the animal pauses for longer times at these locations, then the hippocampus disengages from the current situation and the subsequently emerging sharp wave ripples replay more remote experiences, mixed with the current one. In these late-occurring events, the deep-layer neurons of the entorhinal cortex are also recruited. In

To translate plans into actions, the hippocampus needs to shift from a sharp wave state to a theta oscillation state, during which the subconsciously “primed” circuits can be deployed more efficiently, with goal-contingent valuation of the simulated outcomes occurring in areas downstream of the hippocampal system, likely including the lateral septum–hypothalamus action path or the prefrontal cortex, orbitofrontal cortex, and basal ganglia.³¹

In the preceding four chapters and especially in Chapter 5, I argued that multiple episodes of experience or perhaps multiple sharp wave ripples are needed to generate abstract, semantic information because only multiple experience allows the spatial-temporal context of episodes to be stripped off, leaving behind explicit information. In the next chapter, I explain that acquisition of semantic knowledge can be accelerated by externalizing brain functions and that such externalized functions in the form of artifacts, language, and other forms of communication. Such externalization has been key to the exponential growth of humankind’s knowledge.

SUMMARY

The sharp wave ripple in the hippocampus is the most synchronous population event in the mammalian brain. Ripple events can play palindrome, so that place cell sequences prior to choosing a particular path are replayed in the same order as during crossing the path, only much faster. At the end of the travel path, the same sequence is replayed but now backwards, as if the brain recapitulated a virtual reversed run. Thus, ripple sequence events represent mental travel both into the future and the past.

There are at least two functions of the forward and reversed cell assembly sequences. First, such sequences reach back to the past to replay snippets of waking experience at times when the brain is disengaged from the outside world. This process may consolidate episodic memories and stitch together discontinuous experiences, thereby giving rise to creative thoughts. Second,

contrast, superficial-layer entorhinal neurons, the inputs to the hippocampus, appear to replay sequences largely independent from those in the hippocampus (O’Neill et al., 2017).

31. Priming is an example of what is often referred to as implicit memory (Graf and Schacter, 1985) or “level of processing” (Craik and Tulving, 1975). Redish (2016) discusses extensively how the hippocampal system can cooperate with the networks of the basal ganglia to perform planned actions. Tingley and Buzsáki (2018) demonstrated how the spatial map of the hippocampus is converted into a firing rate-independent, theta oscillation-dependent phase code in the lateral septum, an important conduit from the hippocampal system to the motivational and movement controlling circuits of the brain.

the forward sequences of neuronal assemblies can be viewed as internalized vicarious trial-and-error mechanisms, which can assist with subconscious optimization of future plans. Because the same neuronal substrate can perform both retrospective and prospective operations, the traditional separation of postdiction (i.e., memory) from prediction (i.e., planning) needs to be readdressed (see Chapter 10).

Studying the mechanisms and behavioral relevance of sharp wave ripples raises a deeper problem, which is the “causal” direction (Chapter 2) of the relationship between behavior and replay sequences. Is every internally generated sequence imposed on hippocampal circuits by experience (i.e., from outside in)? In that case, the number of sequences should scale with the amount of experience. Alternatively, the extensive CA3 recurrent system can presumably self-generate very large numbers of sequences independent of any prior experience. Under this second scenario, the role of experience is to pick and choose among preexisting sequences, concatenate them, and modify them a bit, but without the danger of destabilizing the balanced dynamic of the brain.

